Example 2.28

If $z=(\cos\theta+i\sin\theta)$, show that $z^n+\frac{1}{z^n}=2\cos n\theta$ and $z^n-\frac{1}{z^n}=2i\sin n\theta$.

Solution

Let $z=(\cos\theta+i\sin\theta)$.

By de Moivre's theorem ,

$$z^n = (\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$$

$$\frac{1}{z^n} = z^{-n}=\cos n\theta-i\sin n\theta$$

Therefore, $z^n+\frac{1}{z^n} = (\cos n\theta+i\sin n\theta)+(\cos n\theta-i\sin n\theta)$

$$z^n+\frac{1}{z^n} = 2\cos n\theta\,.$$

Similarly,

$$z^n-\frac{1}{z^n} = (\cos n\theta+i\sin n\theta)-(\cos n\theta-i\sin n\theta)$$

$$z^n-\frac{1}{z^n} = 2i\sin n\theta\,.$$

Example 2.29

Simplify $\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18}$.

Solution

We have, $\sin\frac{\pi}{6}+i\cos\frac{\pi}{6} = i\left(\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}\right)$.

Raising to the power 18 on both sides gives,

$$\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18} = (i)^{18}\left(\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}\right)^{18}$$

$$= (-1)\left(\cos\frac{18\pi}{6}-i\sin\frac{18\pi}{6}\right)$$

$$= -(\cos 3\pi-i\sin 3\pi)=1+0i\,.$$

Therefore, $\left(\sin\frac{\pi}{6}+i\cos\frac{\pi}{6}\right)^{18} = 1$.

Example 2.30

Simplify $\left(\frac{1+\cos 2\theta+i\sin 2\theta}{1+\cos 2\theta-i\sin 2\theta}\right)^{30}$.

Solution

Let $z = \cos 2\theta+i\sin 2\theta\,.$

As $|z| = |z|^2=z\bar{z}=1$, we get $\bar{z}=\frac{1}{z}=\cos 2\theta-i\sin 2\theta\,.$

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